

HIGH-THROUGHPUT GENOMIC API INGESTION

Abstraction & Computer Science Motivation

Scientific replication depends on frictionless access to raw sequencer outputs. During major health crises, such as the 2014 Ebola Virus epidemic in West Africa, clinical researchers deposit high-density sequences into public repositories like NCBI's Short Read Archive (SRA). For software engineers, SRA ingestion is a complex challenge related to networking as it involves reconciling raw XML schema queries, resolving rate limits/throttles, and managing high-bandwidth parallel downloads across hundreds of sequence files. This project builds a reliable API harvester that automates data querying, parses run metadata on-the-fly, and launches parallel extraction pipelines.

Ingestion Methodology & SRA Ingestion Workflow

The harvester utilizes NCBI's Entrez Direct suite. Rather than using slow manual interfaces, the script executes RESTful queries with *esearch* and streams XML documents directly to *efetch*. *efetch* parses the raw schemas into tab-delimited run metadata matrices. The script extracts specific Run IDs and schedules asynchronous extraction tasks via the SRA Toolkit's *fastq-dump*, dynamically scaling to all available CPU cores.

Ingestion Shell Script Snippet: `fetch Ebola_runs.sh`

```
#!/bin/bash
set -ueo pipefail
BIOPROJECT="PRJNA257197" # Ebola Outbreak Accession ID

# RESTful Query & XML metadata parsing
esearch -db sra -query "$BIOPROJECT" | efetch -format runinfo > runinfo.csv

# Loop through Run IDs and initiate parallel seq extraction
RUN_IDS=$(awk -F',' 'NR > 1 && NR <= 6 {print $1}' runinfo.csv)
for RUN_ID in $RUN_IDS; do
    fastq-dump --split-files --outdir ./raw_seqs "$RUN_ID" &
done
wait
```

Ingestion Metadata Mapping Results

Executing the pipeline on BioProject *PRJNA257197* (Gire et al., Science, 2014) automatically mapped 891 sequence runs. The parsed metadata profiles crucial data attributes—such as sequence length, layout, and sequencing platform—to instruct the cluster how to distribute resource allocation.

SRR1972917	101,000	PAIRED-END	ILLUMINA (MiSeq)
SRR1972918	101,000	PAIRED-END	ILLUMINA (MiSeq)
SRR1972919	101,000	PAIRED-END	ILLUMINA (MiSeq)
SRR1972920	150,000	PAIRED-END	ILLUMINA (NextSeq)
SRR1972921	150,000	PAIRED-END	ILLUMINA (NextSeq)

Technical Takeaway: Paired-End Thread Isolation

A critical parameter used in fastq-dump is *--split-files*. Modern sequencers read DNA from both ends of a fragment, generating complementary forward (Read 1) and reverse (Read 2) mate pairs. This double-read method provides spatial coordinates crucial for resolving repetitive genomic structures. If these reads are merged into a single file, downstream sequence aligners will fail to align the coordinate mappings, creating high error rates. Using *--split-files* isolates the mates into discrete streams (for example, *1.fastq* and *2.fastq*), allowing downstream parallel alignment indexing to proceed cleanly.

Computer Science Takeaway/Significance: Programmatic REST harvesting converts unstructured genomic repositories into clean, well-defined schemas, establishing automated, reproducible pipelines that form the foundation of cloud-based genomics and sequences.